

Real-Time Phase Conditioned Human Motion Forecasting for Wearable Robot Control

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Abstract—Exoskeleton control requires prediction of human motion to ensure effective assistance and adaptation to user intent. Estimating only the current human motion is insufficient; predictive models are needed to compensate for system latency and to enable proactive assistance. While prior methods have explored predictive control, they are computationally expensive, limiting their suitability for real-time deployment. Drawing inspiration from phase-based generative modeling techniques in computer animation, we introduce a framework for real-time human motion prediction. Our approach integrates wearable inertial measurement unit data with a Periodic Autoencoder and a Mixture of Experts future motion predictor to construct a predictive model of human lower-limb motion. We train and evaluate our method on the Georgia Tech Second Skin dataset, which provides synchronized wearable sensor signals and ground-truth inverse kinematics and kinetics. Our results demonstrate that the proposed framework achieves high prediction accuracy over a 0.5 sec prediction horizon. The model attains a RMSE of 0.175 rad with $R^2 > 0.8015$ for joint angle prediction, and a RMSE of 0.185 Nm/kg with $R^2 > 0.7107$ for joint moment prediction. We show that the framework generalizes effectively to unseen subjects and diverse locomotion modes. We implement the model in real-time on a hip-knee exoskeleton, achieving a mean inference time of 3.784 ms and mean end to end latency under 11 ms, demonstrating its ability to predict human joint kinematics and kinetics with low latency. Our results suggest that leveraging these phase-based generative models in combination with wearable sensors can provide robust, adaptable, and computationally efficient solutions for real-time exoskeleton assistance in real-world settings.

I. INTRODUCTION

In wearable robotics, exoskeletons and prostheses operate in continuous physical and dynamical coupling with the user. Accurate prediction of human intent - the user's current state and short term future motion, is essential to enable proactive assistance strategies that enhance safety, efficiency and user acceptance [1], [5]. Unlike traditional robotic systems, wearable devices are directly integrated with the human body, where sensing, computation, and actuation delays introduce inherent system latencies. Real-time human motion prediction enables the system to anticipate gait evolution, prepare for transitions across locomotion modes, and adapt assistance accordingly - capabilities that reactive controllers cannot provide [2]. By embedding real time human motion forecasting within the control pipeline, exoskeletons can deliver seamless, responsive, and task adaptive assistance [5].

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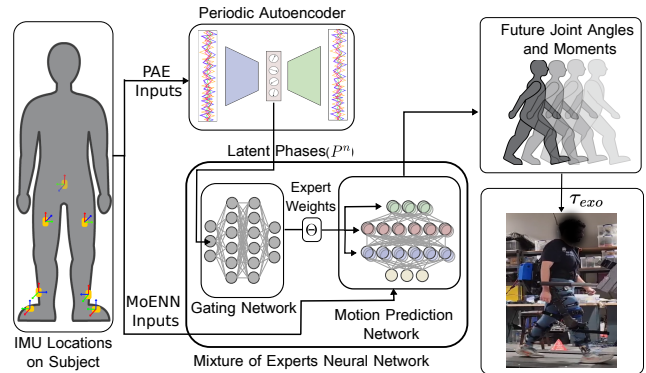


Fig. 1. Overview of the proposed human motion prediction model inspired from computer animation. The method utilizes the data received from the wearable sensors and Phase information from the Periodic Autoencoder to predict future Joint Angles and Joint Moments. These future states act as control inputs to a hip-knee exoskeleton.

However, human motion is complex and modeling it is challenging due to the numerous degrees of freedom and high range of possible motions [2]. Effective prediction, therefore, requires learning both spatial relationships and temporal dynamics. Common approaches for human motion prediction struggle with long-horizon generalization, feed-forward methods capture only local patterns and drift over time, and even recent deep learning models show inconsistent performance across subjects and locomotion modes ([12]–[14]). These limitations underscore the need for more robust approaches to human motion prediction.

Interestingly, the field of computer animation has long confronted the challenge of generating life-like human motion for realistic and believable character movement [8]. Generative models, which utilize latent phase variables to represent motion cycles, have proven capable of producing different contact-aware animations across diverse activities such as walking, running, dancing, playing soccer, etc. ([10], [11]). These models effectively capture both spatial and temporal dynamics by aligning motion prediction to an underlying phase representation. Yet, while these approaches have achieved remarkable success in animation and virtual environments, their application to real-time, sensor-driven motion prediction remains unexplored. This presents a critical gap: can these generative methods, proven in animation, be adapted to noisy sensor data for real-world, real-time human motion prediction for exoskeleton control?

The practical deployment of real-time human motion prediction in wearable robotics requires modeling approaches that remain robust to noisy Inertial Measurement Unit (IMU)

sensors mounted on the body. Although prior work in motion prediction has mainly relied on static, camera-based motion capture systems [7], these systems are confined to lab environments, small capture regions (10m x 10m), and require expensive equipment. In contrast, Inertial Measurement Units (IMUs) provide an inexpensive alternative to motion capture in real-world environments [3]–[5]. However their lower overall accuracy in comparison to marker-based camera systems will require greater care in data processing and motion modeling to get similar results.

To address these challenges, we propose a real-time motion prediction framework operating at 200 Hz that explicitly separates long and short-horizon motion dynamics. To learn long horizon temporal dynamics we utilize a Periodic Autoencoder (PAE) which encodes human motions into a latent space of phase functions. The short horizon components which better reflect the current spatial state of the human motion are represented with a Mixture of Experts style motion prediction Neural Network (MoENN). By employing a mixture of experts network gated by latent phase representations, the proposed methodology achieves high accuracy across multiple prediction horizons, with joint angle RMSE ranging from 0.14–0.185 rad ($R^2 \approx 0.80$) and joint moment RMSE ranging from 0.18–0.22 Nm/kg ($R^2 \approx 0.71$).

To address the aforementioned limitations, we present our key contributions below :

- A real-time phase conditioned motion prediction architecture that decouples long horizon phase dynamics (PAE) from short horizon motion dynamics (MoENN), enabling real-time forecasting from wearable IMUs across subjects and locomotion modes.
- The first adaptation of phase-based generative modeling techniques to noisy wearable IMU signals for robust real-world human motion forecasting.
- Embedded deployment on a hip–knee exoskeleton demonstrating the feasibility of the methodology for prediction-based torque control.

To our knowledge, this is the first phase conditioned latent-space forecasting model driven purely by wearable IMUs that jointly predicts lower-limb kinematics and kinetics and is deployed in real time on an exoskeleton controller.

II. RELATED WORKS

A. Motion Estimation

Optical motion capture, such as Vicon, remains the gold standard for accuracy in motion estimation methods and is widely used as ground truth in datasets like AMASS [15] and the Georgia Tech Second Skin dataset [6]. However, these systems require specialized hardware and controlled laboratory environments, which limits their use in real-world settings. Markerless motion capture approaches reduce the need for markers but remain highly sensitive to environmental factors such as occlusion and lighting.

Wearable sensor systems, particularly IMU-based platforms, provide portability and robustness to environmental conditions. Commercial systems such as Xsens [16],

which use 17 IMUs, achieve high-fidelity full-body tracking through model-based pipelines. More recently, machine learning approaches have outperformed analytical methods by learning temporal dependencies in time-series data. Techniques such as gradient-boosted trees, MLPs, CNNs/TCNNs, RNNs/LSTMs, and Transformers have all been applied to joint state estimation from IMU data ([2], [7]). The Georgia Tech Second Skin [3] exemplifies this direction by combining a wearable IMU suit with a TCNN-style model trained against marker-based ground truth to estimate inverse kinematics and dynamics across multiple locomotion modes. Together, these works demonstrate that wearable sensing, when combined with data-driven modeling, enables accurate and low-latency motion estimation.

Despite these advances, IMU-based systems face drawbacks such as sensor noise, drift, and the need for frequent calibration. Learning-based models may overfit to specific subjects or activities, limiting generalization across populations and locomotion modes. Moreover, most methods estimate only the current state without predicting future motion, leaving a gap between state-of-the-art estimation and the need for real-time forecasting.

B. Motion Prediction

Motion prediction extends estimation by forecasting future human states, enabling a proactive control strategy. Wang et al. [17] proposed an imitation learning framework that combined behavior cloning with generative adversarial imitation learning, while Parsaeifard et al. [18] decomposed prediction into global trajectory and local pose subtasks using LSTM and VAE modules. Wu et al. [19] developed a mixed-reality martial arts training system that used deep learning and RGB cameras to forecast movements for student instruction. While effective, these approaches depend heavily on mounted or portable cameras, which restrict their real-world applicability.

Wearable sensor-based forecasting has emerged as a promising solution. Some methods fuse IMUs and cameras [20] for 3D pose reconstruction, while MoFcNet [5] proposed a two-stage IMU-based pipeline for hip exoskeleton control. However, such multi-stage and temporally deep architectures introduce significant computational overhead, limiting their suitability for embedded deployment in high-frequency assistive control loops. Since exoskeleton systems typically operate at 100–200 Hz, motion prediction must maintain bounded, deterministic inference time. This motivates the development of lightweight forecasting frameworks designed explicitly for real-time hardware deployment.

C. Character Animation

Research in character animation provides complementary approaches to human motion modeling. Traditional pipelines relied on hand-authored key-frames and procedural state machines [23], which achieved real-time performance but struggled with blending different motions, produced artifacts such as foot sliding, and required large datasets

to cover diverse behaviors. Neural motion synthesis approaches—including temporal convolutions, recurrent networks, feed-forward models, and generative methods—scale more effectively but still face issues with ambiguity and temporal misalignment, often resulting in overly smoothed trajectories ([8]–[11]).

Phase-conditioned models such as PFNN [8] align frames using contact-derived global phase variables. Later this idea of global phase was extended to limb-specific phases (local motion phases) for multi-contact behaviors [21]. However, these methods rely on hand-crafted heuristics and break down when motions lack clear contacts. The DeepPhase [11] approach advances this idea by learning multi-channel phase variables through Periodic Autoencoders (PAE), which capture both global and local periodicity directly from motion data. Instead of using labels for contacts, the model builds a multi-dimensional phase manifold where motion trajectories are organized like cycles in polar coordinates, with angle representing progression in time and amplitude encoding motion intensity. This manifold provides smooth temporal alignment across diverse motions, enabling generalization to locomotion, upper-body, and contact-free behaviors

These phase-informed generative models provide robust temporal structure for motion forecasting. We aim to implement a similar strategy on signals received from wearable IMU sensors to learn a latent phase representation to ensure robust motion prediction.

III. METHODOLOGY

We present our proposed framework for human motion prediction, a Mixture-of-Experts style Neural Network (MoENN) containing a gating network informed by the latent phase learned by a periodic autoencoder (PAE), illustrated in Fig. 1. The Methodology section is organized as follows: Section III-A describes the model inputs and outputs, Section III-B outlines the architecture, Section III-C details the dataset and pre-processing steps, Section III-D explains the training procedure, and Section III-E details the real-time implementation and deployment on a hip-knee exoskeleton.

A. System Inputs and Outputs

The time-series model was trained on the Georgia Tech Second Skin dataset [3], which provides time-synchronized IMU data along with ground truth inverse kinematics and kinetics. We used data from seven IMUs placed on the pelvis, thighs, shanks, and feet (Fig. 1).

The gyroscope signals from the IMU are first processed by the PAE to extract the latent phase variables (P^n), which act as one of the inputs to the MoENN. The gating network computes expert weights [11], which, together with the sensor inputs (X_i^M), are used to predict future joint states ($Y_{i+1:i+k}$). Inputs consist of 3 DOF gyroscope and accelerometer signals, and outputs are the ground truth joint angles and moments of the hip, knee, and ankle.

1) *System Inputs:* PAE Inputs:

$$X_i^P = \{X^{\theta}\} \in R^{3N \times t_1} \quad (1)$$

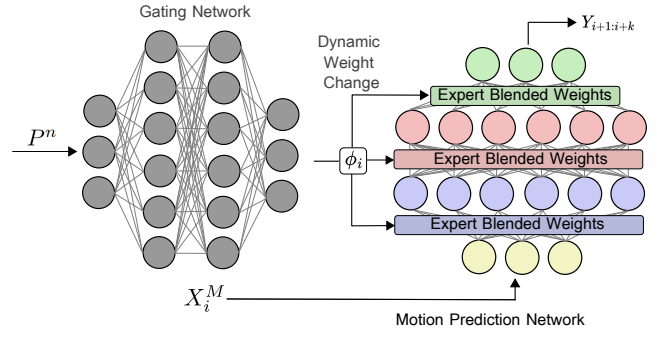


Fig. 2. Detailed view of the Mixture of Experts Neural Network (MoENN). The left Multi-Layer Perceptron corresponds to the Gating Network, right represents the Motion Prediction Network, P^n as Latent Phases from the PAE, X_i^M the wearable sensor inputs and $Y_{i+1:i+k}$ the future predicted joint states.

where $t_1 \in [i - 200, i]$, $N = 7$ number of IMUs and X^{θ} refers to the gyroscope signal about the x,y,z axes of the IMU. The input signal was arranged into channels of overlapping time windows of previous 200 time steps (1.0 sec of past data). The input signals were centered about the mean of each time window.

MoENN inputs:

$$[X_i^M] = \{X^{\theta}, X^{\ddot{q}}, P^n, W, H\} \quad (2)$$

- $X^{\theta} \& X^{\ddot{q}} \in R^{3N \times t_2}$, $t_2 \in [i - 50, i]$: Refers to the respective gyroscope and accelerometer signals from the IMU. The inputs to the MoENN was arranged into 42 channels of overlapping time windows of previous 50 time steps (0.25 secs of past data). All input features were standardized independently across the entire dataset.
- $P^n \in R^{5n}$: Refers to 5 latent phase parameters per phase channel of the PAE.
- W : Subject weight.
- H : Subject height.

2) *System Outputs:* Future joint states

$$[Y_{i+1:i+k}] = \{Y^{\theta}, Y^{\tau}\}, \quad k \in [1, 100], \quad (3)$$

$[Y_{i+1:i+k}]$ consist of ground truth joint angles and moments, where $[i + 1 : i + k]$ where $k \in [1, 100]$, is the prediction horizon specified at the start of training. Y^{θ} and $Y^{\tau} \in R^5$: three degrees of freedom at the hip (flexion-extension, adduction-abduction, rotation), one at the knee (flexion - extension) and ankle (plantarflexion - dorsiflexion).

B. Network Architecture

The motion prediction network is a simple fully connected neural network whose weights are dynamically modulated by a gating network (as shown in Fig. 2). The gating network computes these weights using phase information obtained from the latent channels of the Periodic Autoencoder (PAE).

We first outline the design of the PAE, adapted from the DeepPhase framework [11], which learns a phase manifold from the diverse locomotion patterns in the dataset. We

then present the structure of the Mixture of Experts style Neural Network (MoENN), which leverages this learned phase manifold to predict future joint states.

1) *Periodic Autoencoder*: We implement a PAE based on DeepPhase [11] to learn a phase manifold that captures the periodicity of human motion. The model is trained in an autoencoder framework, reconstructing the input motion while constraining its latent channels to represent periodic functions within a low dimensional latent space.

$$L = g(X_i^P), \quad L \in R^{n \times N} \quad (4)$$

The inputs (X_i^P) are passed through the encoder $g(\cdot)$ that contains multiple 1D convolutional layers to obtain a lower dimensional latent embedding L with $n = 10$ phase channels. Each channel is parameterized as a sinusoidal curve, where amplitude A , frequency F , and offset B estimated via a differentiable FFT layer, and the phase shift S is learned through a fully connected layer to ensure temporal alignment.

$$\hat{L}_i(t) = A_i \sin(2\pi(F_i * t - S_i)) + B_i \quad (5)$$

From these parameters $P^n = A, F, B, S$, the network reconstructs a signal as a set of periodic functions (Eq. 5).

$$\hat{X} = h(\hat{L}) \quad (6)$$

Finally, a decoder ($h(\cdot)$) contains 1D deconvolution layers which maps this latent representation $\hat{L}$ back to the original input signals (Eq. 6). By enforcing the periodic structure in the latent space, the PAE extracts a temporally consistent phase manifold aligned along the diverse locomotion patterns and transitions within the dataset.

2) *Mixture of Experts Neural Network*: We implement a Mixture-of-Experts Neural Network (MoENN) (Fig. 2) to predict future joint states from the IMU signals. The architecture consists of two components:

a. *Gating Network*: This is a 3 layer MLP that takes in the latent phase features P^n from the PAE and produces a set of expert weights. (Eq. 7)

$$\alpha(P^n) = \sigma(W_2 \text{ELU}(W_1 \text{ELU}(W_0(P^n) + b_0) + b_1) + b_2) \quad (7)$$

where W_0, W_1, W_2 and b_0, b_1, b_2 denote the layer weights and biases. The output dimension equals the number of experts (10 in our case). These coefficients define the dynamic weights:

$$\phi_i = \alpha(P^n) \quad (8)$$

b. *Motion Prediction Network*: The prediction network $\beta(\cdot)$ maps sensor inputs X_i^M to future joint states:

$$Y_{i+1:i+k} = \beta(X_i^M; \phi_i), \quad (9)$$

The network consists of a Mixture of Experts (MoE) architecture with three fully connected layers with a hidden size of 256 neurons and ELU activations. Rather than using fixed parameters, each layer's weights are dynamically modulated by the latent phase and expressed as a convex combination

of expert weight matrices. As illustrated in Fig. 2, the gating network modulates these weights using the learned phase manifold to generate future joint states from wearable sensor inputs from the motion prediction network.

C. Dataset

We used the Georgia Tech Second Skin Dataset [3], which provides comprehensive motion data from wearable sensors and ground-truth inverse kinematics and kinetics calculated from time-synchronized VICON motion capture and force plates. The dataset contains recordings from 10 subjects performing multiple trials of various tasks (walking, incline walking, squats, stairs and ladder climbing). Each participant was equipped with IMUs placed on the pelvis, thighs, shanks, and feet. For training, we focused on dynamic tasks including flat-ground walking, incline and decline walking, symmetric and asymmetric squats, and stair and ladder climbing. This dataset was selected because our own sensor suit is modeled after the Georgia Tech Second Skin, allowing us to leverage the dataset for training and directly deploy the learned models to our robotic platform.

D. Network Training

For training our models, data from 9 subjects was used to form the training and validation set, while 1 subject was held out entirely for testing. This ensured that the models never encountered this participant or their conditions during training. The training and validation sets were created by randomly forming subject-condition pairs and splitting them 85% for training and 15% for validation.

Both models were trained separately: the Periodic Autoencoder (PAE) was trained first, and its learned phase representation was then used as input to train the MoENN. The PAE was trained for 200 epochs using the Adam optimizer (learning rate 1×10^{-4}), with 200 frames as input with a batch size of 128. Batch normalization was applied throughout the network to stabilize optimization and reduce overfitting. A mean squared error (MSE) loss was used to train the reconstruction objective.

The MoENN was trained with a shorter input vector of 50 frames (0.25 s). The phase input was obtained from the forward pass of the pretrained PAE encoder and fed into the gating network, which dynamically computed the expert weights of the motion prediction network. The MoENN was trained with the same batch size (128) and optimizer settings (Adam, learning rate 1×10^{-4}).

We chose a mean square error loss (MSE) as our loss function for training the PAE-MoENN:

$$\text{Loss} = \|Y_{i+1:i+k} - \hat{Y}_{i+1:i+k}\|_2^2 \quad (10)$$

In addition to IMU, phase inputs, and subject specific parameters, the MoENN was trained to predict a fixed future horizon k , specified at the start of training. We trained separate models for different prediction horizons ($k = 1, 5, 20, 50, 80$ and 100), covering both short and long-term predictions (from 0.005-0.5 sec).

For reproducibility and implementation details, we provide a link to our GitHub repository in the Appendix section.

E. Real-Time Implementation on a Hip-Knee Exoskeleton

The real-time feasibility of the proposed framework was validated by deploying the model on a hip-knee exoskeleton developed by the Institute for Human and Machine Cognition (IHMC) (Fig. 3), which contains three controlled degrees of freedom: hip abduction-adduction, hip flexion-extension, and knee flexion-extension. A dedicated PAE-MoENN model was trained using pelvis, thigh, and shank IMU measurements to predict future joint angles ($Y_{i+1:i+k}^\theta$) and joint moments ($Y_{i+1:i+k}^\tau$). For deployment, the training data was downsampled from 200 to 100 Hz to match the exoskeleton control frequency. This ensured consistency between the training and deployment conditions and avoided any temporal duration shift due to the end to end latency of sensing, inference, and actuation. The IMUs were mounted at corresponding anatomical locations on the human subject to replicate the dataset conditions (Fig. 3).

The trained network was executed onboard an NVIDIA Jetson Orin Nano for fully embedded inference. The predicted future joint moment was scaled and applied as the assistive control input (Fig. 4) to the exoskeleton:

$$\tau_{exo} = k_\tau(Y_{i+k}^\tau) \quad (11)$$

where k_τ is a manually tuned assistive torque gain.



Fig. 3. Real-Time Deployment Setup. Top Image: Hip-knee exoskeleton with dataset-consistent IMU placement on subject. Bottom Left: Participant performing flat-ground walking. Bottom Right: Participant performing symmetric squats. A supplementary video demonstrating the real-time hardware deployment is provided.

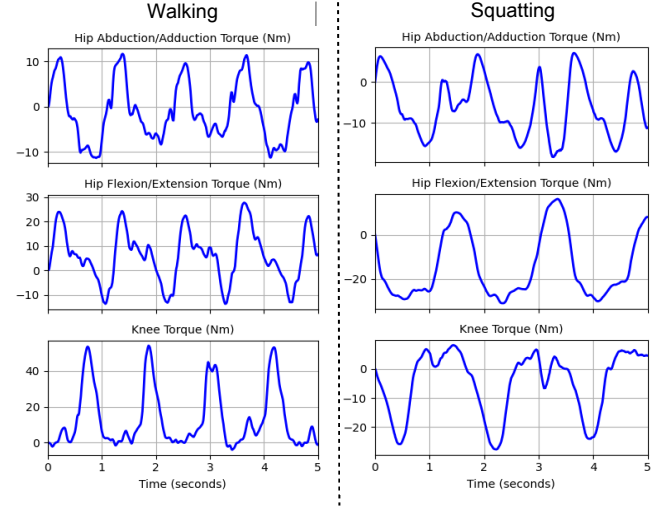


Fig. 4. Real-time predicted joint torque outputs generated online (Y^τ) the hip-knee exoskeleton during walking (left) and squatting (right) across hip abduction/adduction, hip flexion/extension, and knee joints. Signals demonstrate periodic torque profiles produced 100 Hz, confirming the model's ability to generate smooth assistive control inputs in real time.

IV. VALIDATION

We evaluated our model's performance on the one subject held out during training to assess performance across various locomotion modes, joints and different prediction horizons. In addition, we compare the prediction accuracy of our computer animation-inspired PAE-MoENN approach against standard baseline models across different prediction horizons (k). Furthermore, we validate the real-time feasibility on a wearable hip-knee exoskeleton (Fig. 3, Fig 4).

A. Study Setup

As discussed in the previous section, validation was performed using the held-out subject dataset. For comparison, three baseline models were trained under the same conditions: (1) a Fully Connected Neural Network with a Sliding Window input (FCNN-SW), consisting of five linear layers with 512 hidden layer neurons, (2) a recurrent neural network (LSTM) adapted from the work of Fragkiadakis et al. [22], and (3) a Temporal Convolutional Neural Network (TCNN) adapted from the Georgia Tech approach [4] for estimating the joint angles and moments. All baseline models were trained on the same data and under the same conditions as the PAE-MoENN.

B. Results

Table I summarizes the prediction accuracy across various locomotion modes. Joint angle prediction RMSE ranges from 0.12 to 0.20 rad, while joint moment RMSE ranges from 0.18 to 0.38 Nm/kg. Across walking conditions, angle RMSE remains within 0.12 to 0.13 rad, indicating a consistent performance across locomotion modes. Fig. 5 shows the predicted versus the ground truth trajectories for representative locomotion modes.

Table II reports joint-wise accuracy across a prediction horizon of 0.5 sec ($k = 100$). Joint-wise predictions report

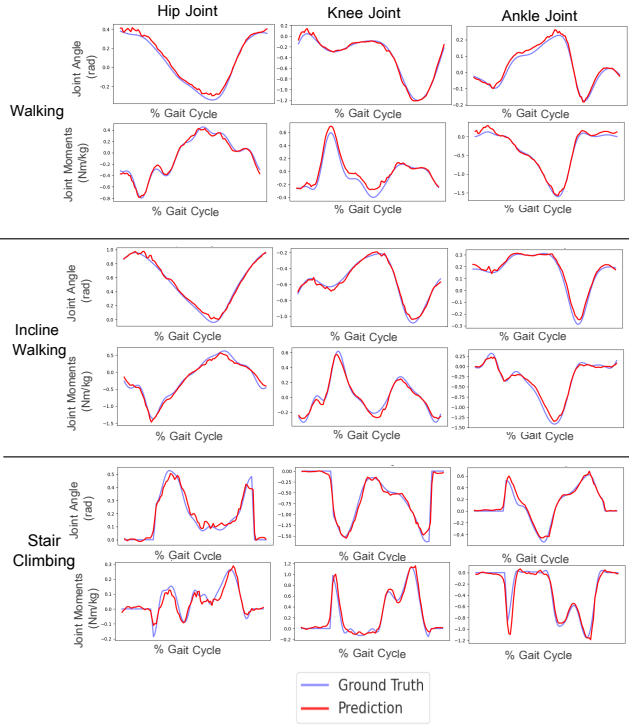


Fig. 5. Comparison of model predictions with ground-truth inverse kinematics for forecasting across multiple locomotion tasks (Flat Ground Walking, Incline walking and Stair Climbing.)

TABLE I

COMPARISON OF MOTION PREDICTION ACCURACY ACROSS DIFFERENT LOCOMOTION MODES

Locomotion Mode	Joint Angles RMSE (rad)	Joint Moments RMSE (Nm/kg)
Flat Ground Walking	0.1291	0.1823
Incline Walking (5 deg)	0.1202	0.2116
Incline Walking (10 deg)	0.1332	0.2186
Decline Walking (5 deg)	0.1260	0.2255
Decline Walking (10 deg)	0.1233	0.2536
Stair Climbing	0.1956	0.2191
Ladder Climbing	0.1595	0.1805
Symmetric Weighted Squat	0.1840	0.3836
Asymmetric Weighted Squat	0.2043	0.2663

a high correlations, with R^2 values ranging from 0.665 to 0.8815 for joint angles and 0.5573 to 0.7684 for moments. RMSE remains low, between 0.10-0.195 rad for joint angles and 0.04-0.23 Nm/kg for moments.

Table III compares various baseline model performance for a prediction horizon of 0.5 sec. The PAE-MoENN achieves the lowest RMSE and highest R^2 for both joint angles (0.1329 rad, $R^2 = 0.8015$) and joint moments (0.1848 Nm/kg, $R^2 = 0.7107$).

In addition, an ablation analysis (Table IV) was conducted to isolate the contributions of the phase representation and expert gating architecture. Removing the phase input or replacing phase-conditioned gating with static expert weights resulted in a consistent degradation in prediction accuracy, particularly at the longer prediction horizon of 0.5 sec.

Fig. 6 illustrates the RMSE as the prediction window increases from 0.005 sec to 0.5 sec, showing that the PAE-MoENN maintains the lowest error across all prediction windows.

The Hardware deployment demonstrated a mean end-to-end latency of 10.335 ms (96.75 Hz). PAE-MoENN inference time averaged out to be 3.784 ms per cycle (36.6% of total loop time), confirming real-time feasibility. Fig. 4 illustrates the predicted joint torque control signals generated onboard during walking and squatting, demonstrating physiologically consistent, repeatable periodic torque profiles across hip and knee joints in real time. These predictions serve directly as assistive control inputs to the exoskeleton, confirming the framework's capability to generate smooth, task-appropriate control signals online. A supplementary video demonstrates real-time execution.

TABLE II

COMPARISON OF PREDICTION ACCURACY ACROSS THE HIP, KNEE AND ANKLE JOINTS OVER A 0.5 SEC PREDICTION HORIZON

Joint Angles (rad)	MAE	RMSE	R^2
Hip Flexion-Extension	0.1157	0.1621	0.8815
Hip Abduction-Adduction	0.0606	0.1242	0.7010
Hip Rotation	0.08335	0.1078	0.814
Knee	0.1304	0.1942	0.7994
Ankle	0.0791	0.1056	0.6640
Joint Moments(Nm/kg)	MAE	RMSE	R^2
Hip Flexion-Extension	0.1840	0.23695	0.7684
Hip Abduction-Adduction	0.1306	0.1850	0.6345
Hip Rotation	0.03255	0.0479	0.55725
Knee	0.1676	0.2252	0.6785
Ankle	0.1331	0.1734	0.676

TABLE III

COMPARISON OF PREDICTION ACCURACY ACROSS VARIOUS MACHINE LEARNING MODELS OVER A 0.5 SEC PREDICTION HORIZON

Model	Joint Angles (rad)		Joint Moments (Nm/kg)	
	RMSE	R^2	RMSE	R^2
FCNN-SW	0.2050	0.5517	0.2775	0.5454
LSTM [22]	0.1784	0.6351	0.2294	0.5344
TCNN [3]	0.1602	0.7584	0.2275	0.6082
PAE-MoENN	0.1329	0.8015	0.1848	0.7107

TABLE IV

ABLATION STUDY OF PHASE CONDITIONED ARCHITECTURE FOR HUMAN MOTION FORECASTING OVER A 0.5 SEC PREDICTION HORIZON

Locomotion Mode	Joint Angles		Joint Moments	
	RMSE (rad)	R^2	RMSE (Nm/kg)	R^2
PAE-MoENN	0.1329	0.8015	0.1848	0.7107
No Phase (Standard MLP)	0.2079	0.6351	0.2896	0.5344
MoE (no Experts Gating)	0.2193	0.7584	0.2775	0.6082

V. DISCUSSION

This work presents the first phase conditioned latent-space forecasting model, driven solely by wearable IMUs, that

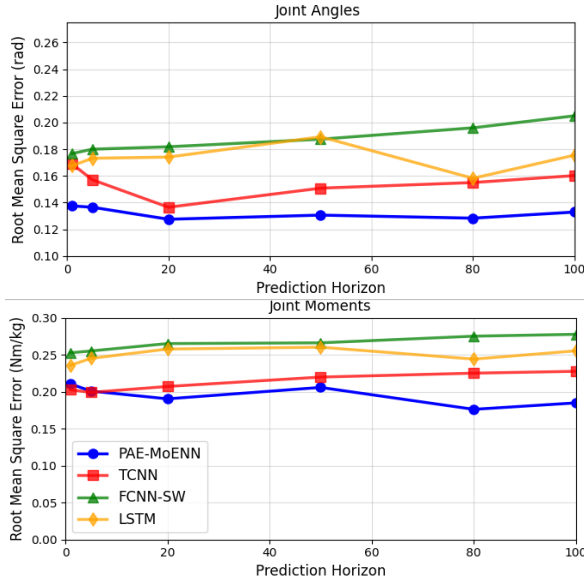


Fig. 6. Comparison of Root Mean Square Error (RMSE) for Joint Angles and Joint Moment Prediction for the four models (FCNN-SW, TCNN, LSTM and PAE-MoENN) across the prediction horizon (0 to 100).

jointly predicts lower-limb kinematics and kinetics and runs in real time on an exoskeleton controller. We evaluated our approach against multiple baseline models across various prediction horizons (Fig. 6), examined its performance across different joints (Table II) and locomotion modes (Table I). It was also tested in real time on a wearable hip-knee exoskeleton, demonstrating its feasibility for online deployment (Fig. 3 and Fig. 4).

By leveraging a phase-conditioned latent space, the model learns a compact representation of the locomotion state and uses that to robustly predict future motion states. Ablation analysis further highlights the importance of this design, removing the phase representation or disabling the phase conditioned expert modulation degrades the long horizon prediction accuracy. This indicates that explicitly encoding the progression of gait phase and modulating expert weights by phase improves alignment across gait cycles and adaptability across locomotion patterns.

Despite these improvements, our method faces several limitations. IMU-based wearable data are inherently noisy and subject to bias drift, complicating accurate joint state estimation. The absence of subject-specific anatomical parameters (e.g., limb lengths, skeletal dimensions) may further limit generalization across individuals. Additionally, the model was trained on a modest dataset with limited subject diversity, which constrains its generalizability to broader populations and movement styles. Additionally, the Georgia Tech Second Skin dataset records locomotion modes in isolation, without natural task transitions. As phase-based generative models are particularly suited to capturing such transitions, future work will target continuous, transition-rich data collection to fully exploit the model’s capabilities.

To overcome these constraints, we developed a wearable sensor suit (Fig. 3) modeled after The Second Skin [3], capa-

ble of capturing continuous real-world data across multiple tasks, transitions, and environments. Beyond richer transition coverage, this hardware enables large-scale data collection across a more diverse subject pool, directly addressing the dataset size and diversity limitations noted above, and allowing us to fully exploit phase-based dynamics for more robust long-horizon forecasting.

While the exoskeleton test demonstrated real-time execution, future work is needed to determine the optimal prediction horizon for exoskeleton control and to maximize the benefits of proactive, anticipatory assistance strategies. Looking forward, phase-conditioned forecasting could be integrated into closed-loop assistive systems, complementing reinforcement and imitation learning to enhance stability, coordination, and adaptability. Future work includes determining optimal prediction horizons for exoskeleton control, leveraging the method for fall detection and balance assessment, and exploring synergistic effects with other learning paradigms. Together, these developments lay the foundation for predictive, adaptive assistive devices that proactively respond to human motion, maximizing biomechanical benefit and user safety.

VI. CONCLUSION

In this work, we introduced a novel phase-conditioned motion prediction framework for wearable assistive devices, inspired by generative models from computer animation (DeepPhase). Leveraging wearable IMU sensor data, the method demonstrated good prediction accuracy across multiple tasks and subjects and outperformed other baseline methods across different prediction horizons. Real-time deployment on a hip-knee exoskeleton further demonstrated the feasibility of online inference, highlighting the potential for proactive assistive control.

Future work will focus on determining optimal prediction horizons for real-time exoskeleton adaptation, integrating richer datasets with continuous task transitions, and incorporating subject-specific anatomical parameters to further improve performance. By combining predictive modeling with adaptive assistive control, this framework offers a promising path toward enhancing safety, coordination, and biomechanical benefit in wearable robotics.

APPENDIX

To preserve anonymity during the double-blind review process, links to supplementary documentation and code are withheld. These materials will be made publicly available upon publication.

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